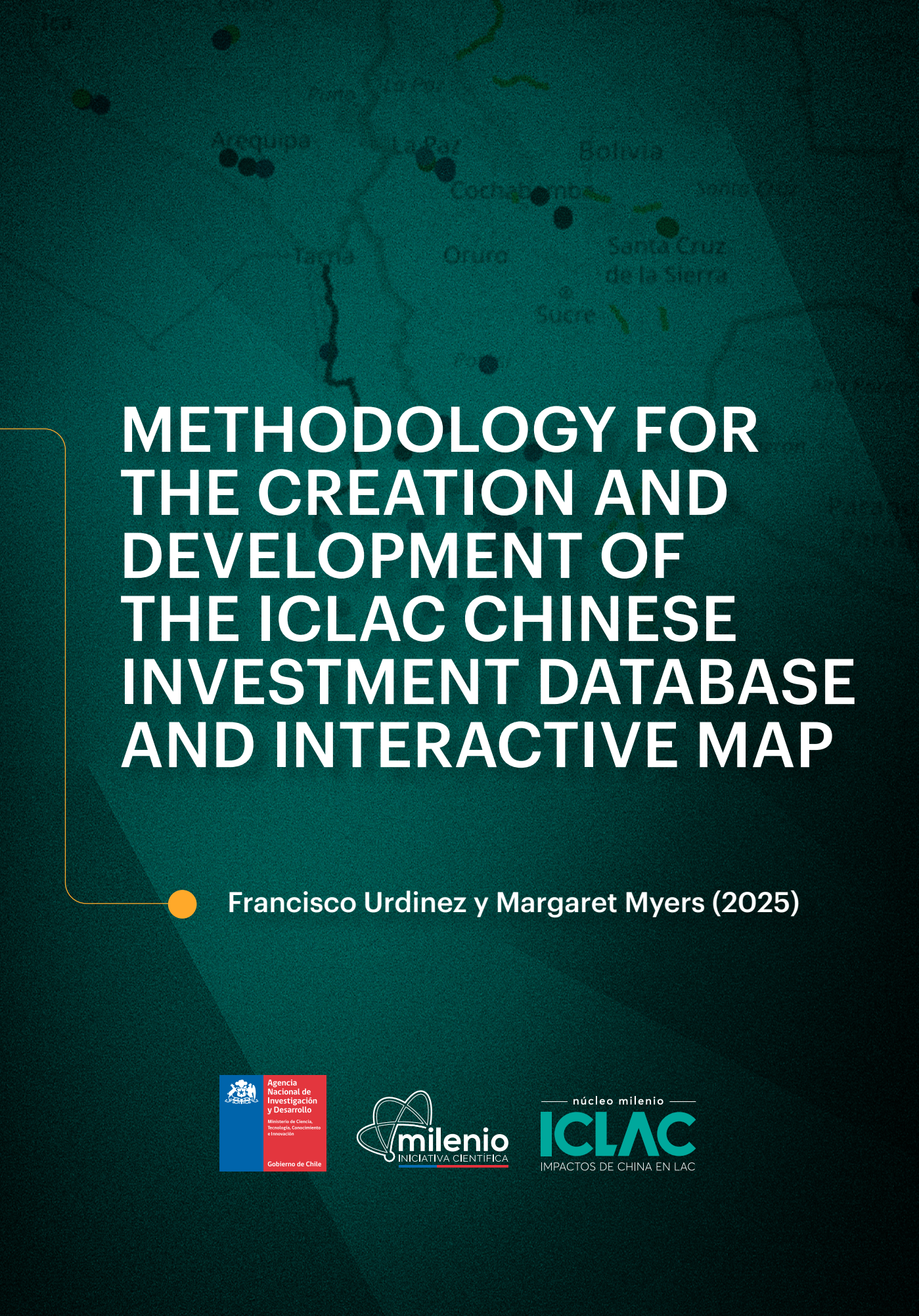




METHODOLOGY FOR THE CREATION AND DEVELOPMENT OF THE ICLAC CHINESE INVESTMENT DATABASE AND INTERACTIVE MAP

Francisco Urdinez y Margaret Myers (2025)



Introduction

A fundamental challenge in documenting Chinese overseas investment stems from the absence of comprehensive, transparent, enterprise-level data from official sources in the People's Republic of China. While the Ministry of Commerce (MOFCOM) publishes aggregate statistics on China's outward foreign direct investment through its annual Statistical Bulletin, these reports provide only country-level totals without identifying specific companies, projects, or investment locations within recipient nations.

This lack of granular official data—which contrasts sharply with the detailed investment registries maintained by many Western countries—creates a significant information gap for researchers, policymakers, and communities seeking to understand the actual scope and nature of Chinese economic engagement in their regions. The opacity of official Chinese investment data has necessitated reliance on secondary sources, including media reports, commercial databases, and host country records, each of which captures only partial aspects of Chinese investment activity.

This data limitation motivated the creation of the ICLAC Regional Repository of Chinese Investments in Latin America, which seeks to construct the most complete picture possible of Chinese investment in Latin America and the Caribbean by systematically triangulating information from multiple primary and secondary sources and subjecting all data to rigorous verification processes.

This methodology report details the systematic six-step process employed to develop, verify, and publish this resource, which represents one of the most thorough documentation efforts of Chinese economic engagement with the region to date.



STEP 1: Initial Database Construction through Media Analysis (2021-2023)

The foundation of this project began with the creation of a comprehensive database documenting Chinese companies with presence in Latin American and Caribbean countries since 2001. This initial phase, funded by ANID Chile (grants 11180081 and 1230307) employed a rigorous manual analysis approach conducted between 2021 and 2023. A research team comprising five research assistants systematically examined the major newspapers in each country, acquiring subscriptions to access their complete archives.

The methodology involved searching newspaper archives using specific keywords including “China,” “investment,” “Chinese company,” “Chinese,” and “Chinese project” in the respective languages. The scope of this media analysis was extensive, covering major publications across the region. In Argentina, for instance, the team examined Clarín, La Nación, and Página 12, while in Brazil, they analyzed Folha de São Paulo, O Globo, and Estadão. Similar comprehensive coverage was achieved across all countries, from smaller Caribbean nations such as Antigua and Barbuda’s Daily Observer to larger markets including Mexico’s El Universal, Milenio, and Excelsior. Additionally, the research incorporated international financial publications including the Financial Times, New York Times, Bloomberg, Wall Street Journal, Xinhua, and South China Morning Post to capture projects that may have received limited local coverage.

Following the identification of preliminary news items, the team conducted supplementary Google searches for each identified project to gather additional information and verification. This extensive process resulted in an original database containing over 4,800 news stories about Chinese investment, sourced from more than 800 newspapers and news portals covering the period from 2001 to 2020.

Each project underwent careful categorization, distinguishing between existing operational investments and those that were announced but never materialized. The coding system included unique project identifiers, investment amounts in US dollars, ownership structure (state-owned versus private enterprises), investment motivation, announcement year, investment

type (greenfield or merger and acquisition), joint ventures, projects location, and industry classification.

This initial database identified a total of 561 investment projects across 23 countries, forming the empirical foundation for Chapter 3 of the book “Economic Displacement” published by Cambridge University Press in 2025 (Urdinez 2025).



STEP 2: Cross-Verification with Authoritative Sources (2022-2023)

The second phase involved meticulous verification of each news story against investment data from established authoritative sources. This triangulation process utilized both subscription-based databases, including fDi Markets and Mergr, and open-access resources such as the China Global Investment Tracker and the China Foreign Direct Investment Observatory in LAC. Each source maintained different levels of precision and focus areas. The China Global Investment Tracker, by design, excludes investments below 100 million USD, while fDi Markets specializes exclusively in greenfield FDI projects, though not exclusively deals struck by Chinese companies. Mergr focuses solely on mergers and acquisitions with limited data on Chinese enterprises, and the China Foreign Direct Investment Observatory in LAC, directed by economist Enrique Dussel Peters, showed significant overlap with fDi Markets data while also incorporating construction projects wherein China retains no ownership.

Where available, the team incorporated reliable national sources to enhance verification. In Chile, for example, InvestChile, a government entity responsible for registering investment projects and attracting foreign investment, provided valuable information about Chinese projects in the country. Similarly, in Brazil, the Brazil-China Business Council and the BRICS Policy Center think tank offered detailed information about Chinese investments in Brazil.

Our project follows the OECD's definition of FDI. At its core, the OECD considers FDI to exist when an investor resident in one country establishes a lasting interest in an enterprise operating in another country. Notably, not all international investment qualifies as FDI. The OECD establishes a crucial threshold: the investor must own at least 10% of the voting power in the foreign enterprise. This percentage isn't arbitrary - it reflects the idea that with 10% or more, the investor has sufficient influence to participate effectively in the management of the company (OECD 2008).

To facilitate the identification of Chinese ownership, we created three ownership levels, namely: full acquisition (100%), majority acquisition (50% - 99%), and minority acquisition (<49%).

During this phase, it became necessary to distinguish a new category of projects that do not qualify as foreign direct investment, namely public works contracts where Chinese companies execute infrastructure projects such as bridges and dams. Many of these projects, which received substantial media coverage, were documented in the AidData China database managed by William & Mary University. Although these projects do not fit the OECD definition of FDI, we decided to retain these projects and identify them as "construction" since they represent a considerable portion of Chinese activity in Latin America and the Caribbean. The repository's construction projects are those wherein Chinese companies have developing projects and being paid for this work, but do not retain ownership of the infrastructure they build, which belongs to the state or a private contractor. We remove construction project values from our counts of total Chinese operational FDI in the region.

In this phase, investment project sectors were standardized into seven categories plus the aforementioned infrastructure projects. These are: energy, manufacturing, real estate, mining, ICT, agribusiness, and finance. Assigning projects to one of these sectors can be challenging, since there are cases that could belong to two sectors. For example, a soy pellet factory could belong to either agribusiness or manufacturing. China's limited forestry investments in the region do not fall neatly into any of these categories, so have been classified as manufacturing given the end use of the lumber in question. Furthermore, within the same category there can be very different projects - for example, the energy category features both solar park and oil drilling projects.

The criterion for defining the investment year was the commencement of construction work, or alternatively, the year when the acquisition operation was completed. In other words, we carefully accounted for the time lag that can exist between the initial announcement and the project's completion. We also considered that many projects are announced but never completed. If a company acquires a project and years later reinvests to expand or improve that project, these are counted as two separate investments and indicated as "second investment" or "second reinvestment" in the repository. If Chinese investment eventually ends because investors sell their assets to another company, this project is classified as terminated and removed from the repository. We maintain a separate, yet-to be published database of projects that were announced but never completed, and of projects that were completed but left the market years later.

This verification stage resulted in the elimination of projects that could not be confirmed by additional sources and lacked either a verified postal address or an official website confirming the investment's implementation. When discrepancies arose between the collected data and these authoritative sources, the team carefully reviewed and corrected investment attributes such as amounts and announcement years.

This phase was conducted with support from a six-person team affiliated with the Núcleo Milenio project on China's Impacts in Latin America and the Caribbean, funded by ANID Chile. Here, each project underwent a multi-stage and multilingual review process during which several adjunct researchers and research assistants from across the region and Asia sequentially scrutinized the dataset: once one researcher completed their verification, the project entry was passed on to another for cross-checking. This serial procedure ensured maximum attention to detail, comprehensive triangulation across multiple sources, and the consistency of coding decisions.

This database was used in chapters 1 and 2 of Urdinez's book (2025). To provide context for the comprehensiveness and reliability of our methodology, we conducted a systematic comparison of our findings with other major databases tracking Chinese investment in Latin America and the Caribbean for the period 2001-2020.

Our methodology at this stage identified a total Chinese investment stock of US\$265.59 billion across the region during this twenty-year period.

This figure represents substantially more investment activity than what was captured by other leading databases. The combined data from fDi Markets and Mergr, two prominent commercial investment tracking services, recorded only US\$140.2 billion in Chinese investment for the same period—approximately 53% of what our comprehensive methodology uncovered. This significant difference can be attributed to our inclusion of smaller-scale investments that fall below the reporting thresholds of commercial databases, as well as our systematic capture of projects that received primarily local media coverage rather than international financial press attention.

The China Global Investment Tracker, maintained by the American Enterprise Institute and Heritage Foundation (Scissors 2021), documented US\$190.01 billion in Chinese investment stock. While this represents a more complete picture than the commercial databases, it still accounts for only 71% of the investments we identified. This gap is primarily explained by the China Global Investment Tracker's deliberate exclusion of all investments below US\$100 million, which our research shows constitute a significant portion of Chinese economic engagement in the region, particularly in manufacturing and agribusiness sectors.

Perhaps most surprisingly, our database captured less than half of the investment reported in the Statistical Bulletin of the People's Republic of China, which claimed US\$629.81 billion in Chinese investment stock in Latin America and the Caribbean. This substantial discrepancy of more than US\$364 billion raises important questions about methodology and definitions. The official Chinese statistics appear to include financial flows and transactions that do not meet the OECD definition of foreign direct investment, such as short-term loans, portfolio investments, or financial instruments that do not establish lasting management interest in local enterprises.



STEP 3: Case Study Identification and Knowledge Gap Analysis (2023-2025)

In partnership with the Inter-American Dialogue, under the leadership of Margaret Myers, the team initiated a systematic search for case studies corresponding to each investment project, aiming to identify knowledge gaps regarding Chinese projects. A qualifying case study was defined as a document in English, Portuguese, Spanish, or Chinese that analyzed a specific investment, contained at least one page of qualitative content about the investment, was published by an academic affiliated with an academic institution, and resulted from academic research.

This phase proceeded in stages, beginning with Chile, Peru, and Argentina in late 2023, followed by Bolivia, Ecuador, Uruguay, Brazil, and Paraguay in early 2024. In 2025, the process expanded to include Colombia, Venezuela, Guyana, Suriname, and Panama. This systematic approach to identifying existing scholarship revealed significant gaps in academic documentation of Chinese investments across the region.

Of the first eight countries published by ICLAC, out of the 363 investment and construction projects represented in our database, only 113 (31 percent) included case studies.



STEP 4: Local Expert Validation (2023-2024)

Following the case study identification phase, the advanced database underwent verification by local partners with expertise in Chinese investment. In Chile, where the project is based, researchers affiliated with ICLAC from Universidad de Chile and Pontificia Universidad Católica de Chile reviewed the database, with additional consultations conducted with the Chile-China Chamber of Commerce. Data on Argentina was reviewed by experts at the China-Argentina Center at the University of Buenos Aires and officials from the Argentina-China Chamber of Commerce. In Peru, experts from the Center for Studies on China and Asia-Pacific (CECHAP) at Universidad del Pacífico conducted the review.

For Brazil, an independent consultant specializing in Chinese investment reviewed the database. In Uruguay, an academic from ORT University provided verification, while in Colombia, the review involved an academic from Universidad del Rosario and members of the Colombia-China Chamber of Commerce. At this stage, the ICLAC database often proved more comprehensive and accurate than the databases maintained by local partners themselves, though the process served to identify new projects requiring addition and to verify investment amounts and project details where uncertainties remained. These regional institutions are critical partners in this project and will continue to work alongside the ICLAC team to verify and update data. These “on-the-ground” partnerships and our collaborative approach importantly distinguish the repository from other efforts to track Chinese investment in the region.



STEP 5: Georeferencing and Spatial Documentation (2024)

With the database from steps three and four completed for eight countries, the project advanced to the georeferencing phase. ICLAC's efforts to georeference instances of operational Chinese investment in the region are intended to convey the geographical ambit of these deals, as well as the extent to which some projects may be interoperable, including across borders. These efforts distinguish the repository from other FDI tracking projects.

The projects in the database were first differentiated between those requiring point mapping (such as factories or offices) and those requiring vector mapping (such as electrical networks, roads, rail, or gas pipelines). Throughout 2024, a team of geographers specializing in GIS systems conducted this work.

To confirm exact project locations, the team consulted official company websites, government documentation, and ministry of public works records for infrastructure projects. For example, Joyvio's salmon farms in southern Chile were identified through company reports filed with the Ministry of Environment. By this verification level, nearly all projects possessed some form of public registration, such as a tax identification number. Each investment was assigned specific coordinates, and for projects conveyed as vectors, coordinates along the project's path were identified and then displayed on the repository's map.



STEP 6: Publication and Interactive Platform Development

The final process involved generation and publication of the interactive map and related functions on the website. These functionalities were developed by the Millennium Institute for Foundational Research on Data (IMFD), which, with input from ICLAC leadership and affiliates, transformed the raw data into an interactive tool and incorporated English, Spanish, and Chinese versions of the web interface. In collaboration with IMFD, the data continues to undergo biannual updates.

The platform is intended to be interactive and includes a contact mechanism allowing users to report errors or provide feedback, ensuring continuous improvement and accuracy of the database. To ensure maximum transparency and also access to and use of the data, raw data is fully available for public download. The platform additionally includes a contact mechanism allowing users to report errors or provide feedback, ensuring continuous improvement and accuracy of the database. From this study, a report was published in English and in Chinese (Urdinez & Myers 2025).

Conclusion

This six-step methodology represents a comprehensive approach to documenting Chinese investment in Latin America and the Caribbean, combining extensive media analysis, multiple verification layers, academic validation, and sophisticated spatial documentation. The resulting database and interactive map provide researchers, policymakers, and stakeholders with an unprecedented resource for understanding the scope, distribution, and characteristics of Chinese economic engagement in the region. The iterative nature of the methodology, with its multiple validation stages, including with repository partner organizations, and ongoing updates, ensures both accuracy and relevance as Chinese investment patterns continue to evolve across Latin America and the Caribbean.

References

- AidData. (2024). *AidData China Database*. William & Mary University. Available at: <https://www.aiddata.org>
- Brazil-China Business Council. (2024). *Chinese Investments in Brazil Database*. São Paulo: CEBC.
- BRICSPolicy Center. (2024). *China-Brazil Economic Relations Database*. Rio de Janeiro: BPC.
- Urdinez, F. (2025). *Economic Displacement: China and the End of US Primacy in Latin America*. Cambridge: CUP.
- Dussel Peters, E. (Ed.). (2024). *China Foreign Direct Investment Observatory in LAC*. Mexico City: UNAM.
- fDi Markets. (2024). *fDi Markets Database*. Financial Times Ltd. Available at: <https://www.fdimarkets.com>
- Inter-American Dialogue. (2024). *China-Latin America Report*. Washington, DC: IAD.
- InvestChile. (2024). *Foreign Investment Projects Database*. Santiago: CORFO.
- Mergr. (2024). *M&A Database*. Available at: <https://mergr.com>
- Millennium Institute for Foundational Research on Data (IMFD). (2024). *Interactive Mapping Platform Development*. Santiago: IMFD.
- Núcleo Milenio. (2024). *Impacts of China in Latin America and the Caribbean*. ANID Chile Research Grant.
- OECD (2008), *OECD Benchmark Definition of Foreign Direct Investment*, Fourth Edition, OECD Publishing, Paris. Available at: <https://www.oecd.org/investment/fdibenchmarkdefinition.htm>
- Mergr M&A database (<https://mergr.com/>)
- Scissors, D. (2021). *China Global Investment Tracker*. American Enterprise Institute. Available at: www.aei.org/china-global-investment-tracker/
- Ministry of Commerce of the People's Republic of China (MOFCOM). (2021). *Statistical Bulletin of China's Outward Foreign Direct Investment 2020*. Available at: <https://fdi.mofcom.gov.cn/resource/pdf/2020/12/09/048b1af24cde4c1995f2245c2a735109.pdf>
- Urdinez, F., & Myers, M. (2025). *Trends in Chinese FDI in South America: Findings from the regional repository of Chinese investments in Latin America*. ICLAC. DOI: 10.5281/zenodo.15080442

List of individual collaborators in alphabetical order

- Allison Stronach (ICLAC)
- Benjamín Muñoz (ICLAC)
- Benjamín Zúñiga (ICLAC)
- Brenda Vladisauskas (CEACH, Universidad de Buenos Aires)
- Cámara Argentino China
- Cámara Colombo China (CCCHIC)
- Camilo DeFelipe (Pontificia Universidad Javeriana)
- Catalina Leiva (ICLAC)
- Centro de Estudios sobre China y Asia-Pacífico (CECHAP), Universidad del Pacífico, Peru.
- Dafne Estesso (CEACH, Universidad de Buenos Aires)
- Darissa Cama (CECHAP)
- Eugenio Sánchez (ICLAC)
- Florencia Muñoz (ICLAC)
- Francisco Urdinez (ICLAC)
- Fundación Andrés Bello, Colombia.
- Gabriela Contreras (IMFD)
- Gonzalo Concha (ICLAC)
- Ignacio Tornero (ICLAC)
- Jessie Hu (Inter-American Dialogue)
- Juliana González Jauregui (ICLAC)
- Margaret Myers (ICLAC)
- Xinyi Wang (Inter-American Dialogue)
- Yifang Wang (Inter-American Dialogue)

COLABORADORES



CENTRO DE ESTUDIOS
SOBRE CHINA Y
ASIA-PACÍFICO



UNIVERSIDAD
DEL PACÍFICO

